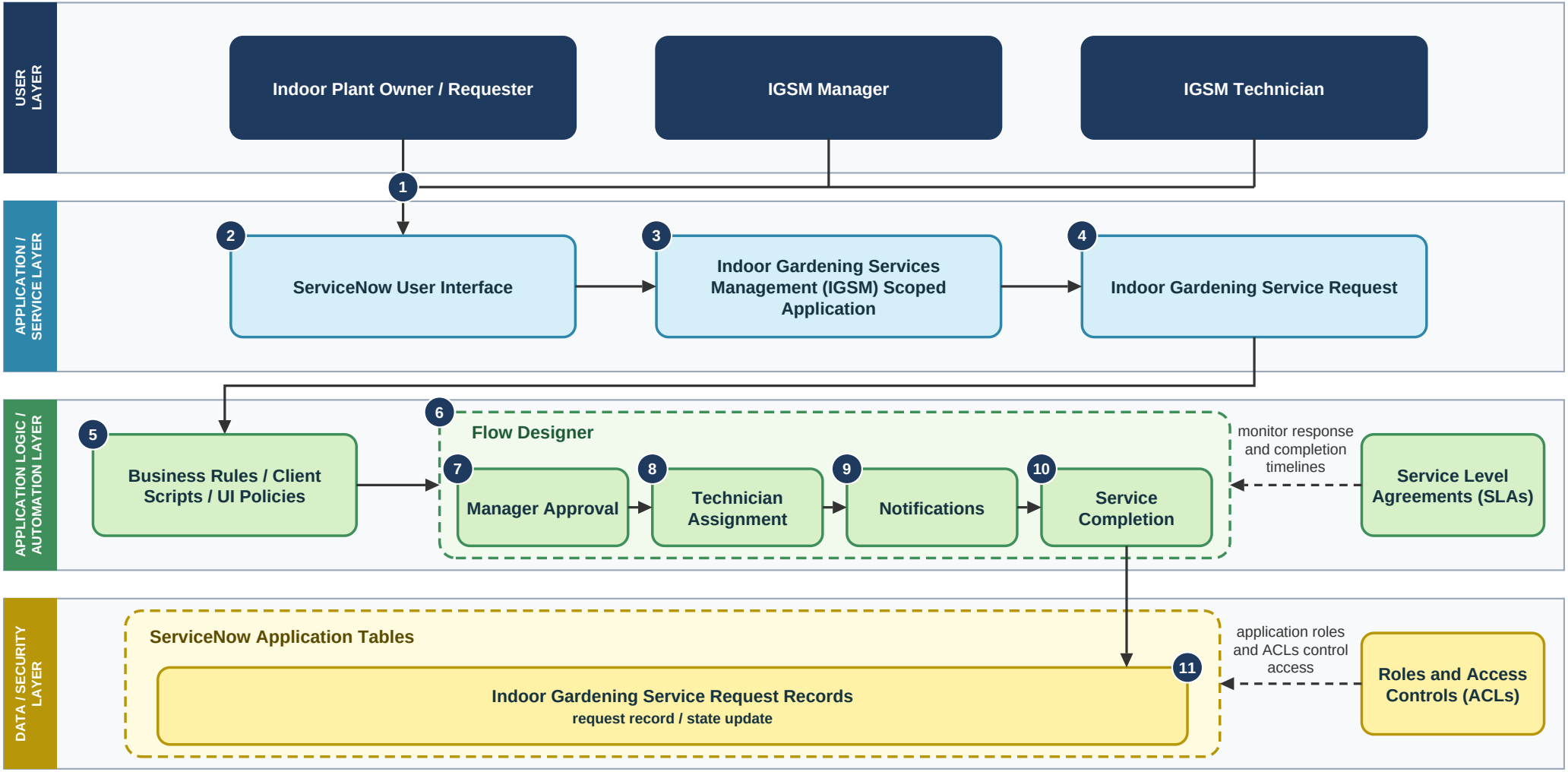
PROJECT DESIGN PHASE-II

TECHNOLOGY STACK

(ARCHITECTURE & STACK)

Project Name	Indoor Gardening Services Management (IGSM)	Project Type	Individual Project

1. Technical Architecture



Numbers 1–11 show the main request flow. Solid arrows show the flow of the service request; dashed arrows show SLA monitoring and access control.

2. Table-1: Components & Technologies

S.No	Component	Description	Technology
1	**User Interface**	Interface through which indoor plant owners, managers, and technicians interact with the IGSM application and service requests.	ServiceNow User Interface / Service Portal
2	**Scoped Application**	Application layer containing the Indoor Gardening Services Management functionality and configuration.	ServiceNow Scoped Application
3	**Application Logic**	Backend logic used to validate request data, apply business rules, and control request processing.	ServiceNow Business Rules
4	**Client-side Logic**	Client-side behavior used to control form behavior and make fields mandatory or dynamically respond to user selections.	ServiceNow Client Scripts and UI Policies
5	**Workflow / Process Automation**	Automates approval, technician assignment, notifications, and request state transitions.	ServiceNow Flow Designer
6	**Database**	Stores indoor gardening service request records and related application data.	ServiceNow Platform Database / Application Tables
7	**Access Control**	Controls access to application records and operations based on user roles and permissions.	ServiceNow Roles and ACLs
8	**Service Level Management**	Tracks response and completion time requirements for gardening service requests.	ServiceNow SLA Definitions
9	**Notifications**	Provides updates to users about important changes in their gardening service requests.	ServiceNow Notifications
10	**Infrastructure / Platform**	Provides the platform environment in which the scoped application, workflows, data, security, and other configurations operate.	ServiceNow Platform

3. Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	**Open-Source Frameworks**	No separate open-source application framework is used. The application is developed within the ServiceNow scoped application environment.	ServiceNow Platform
2	**Security**	Access to application data and operations is controlled using ServiceNow application roles and Access Control Lists (ACLs). Different access levels are provided for requesters, technicians, and managers according to their responsibilities.	ServiceNow Roles and ACLs
3	**Scalable Architecture**	The application is developed on the ServiceNow platform, where application functionality, data, workflow automation, and access control are managed within the platform. The scoped application structure keeps IGSM functionality organized within its application scope.	ServiceNow Scoped Application
4	**Availability**	The application operates within the ServiceNow platform environment, allowing authorized users to access the application and manage service requests through the platform.	ServiceNow Platform
5	**Performance**	The application uses automated Flow Designer processes, business rules, and client-side controls to manage service requests efficiently. The design avoids unnecessary processing and keeps request handling organized.	ServiceNow Flow Designer, Business Rules, Client Scripts and UI Policies

4. Architecture Explanation

Indoor Gardening Services Management (IGSM) is implemented as a ServiceNow scoped application. Users interact with the application through the ServiceNow interface. Service requests are stored in application tables and processed using ServiceNow business rules, client-side controls, Flow Designer, notifications, and SLAs. Roles and ACLs provide controlled access to application data and operations. This architecture supports organized service request management from request creation through approval, technician assignment, notification, and completion.

5. Technology Summary

Technology / Component	Purpose
ServiceNow Scoped Application	Main application platform
ServiceNow Tables	Data storage
Flow Designer	Process automation
Business Rules	Backend validation and logic
Client Scripts / UI Policies	Client-side form behavior
Roles / ACLs	Security and access control
Notifications	User communication
SLAs	Response and completion tracking